S4 Developmental Interpretation

A cautious reader-facing summary

Article type: Supplement

Short title: APPLESEED developmental scope

Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

Scope

Research reanalysis, not a clinical test

This is a secondary analysis of a small public infant EEG example dataset. It tested no intervention, provides no diagnostic threshold, and makes no individual-level prediction. No causal developmental mechanism is established.

This supplement translates the analysis boundary rather than amplifying the findings. The participants are not a clinical cohort, and the NDT-derived summaries are research measurements without clinical validation.

What was studied

Thirteen infants each had two recordings approximately one week apart at four months. Eleven returned at approximately eight and twelve months. The analysis asked whether session-level local-flow and reachability summaries were stable in the short interval and changed within the same children at later assessments.

What the measures mean

The MNJ is a local mathematical summary of how the exported EEG feature chart changes close to a given chart location. Reachability summarizes the local variation predicted by a fitted short-horizon transition model. Neither quantity is a direct measure of consciousness, cognition, disease, maturity, or prognosis.

Main findings

Test–retest agreement was endpoint-dependent and imprecise in this 13-child sample: some point estimates were moderate or high, but the bootstrap intervals were broad and often included zero. No primary 3D MNJ/reachability contrast at eight or twelve months survived its nine-endpoint false-discovery-rate sensitivity correction in the 11-child longitudinal panel. Separately, the cohort-anchored mobility coordinate decreased in all 11 children from the four-month mean to twelve months (raw $p=0.00098$; BH $p=0.0029$ within the three-coordinate family). This is a chart-level common-scale result with moderate coordinate reliability, not evidence for a latent neural or maturational mechanism.

What was not found

The study found no corrected primary MNJ/reachability endpoint difference over time. It did not test diagnostic separation, treatment response, clinical outcome prediction, or a link between the features and behavior. The mobility result should not be read as a normative group trajectory: the global IQR anchor includes all study ages, and nominal age strata are not individually measured ages.

Appropriate next step

A confirmatory developmental study should prespecify a small endpoint set, use a larger longitudinal cohort, retain repeated recordings per age, test anchor stability across visits, and relate any reliable measure to independent behavioral/developmental outcomes.